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# Kosmische Manual



Kosmische generates MIDI sequences in the style of the Berlin School (Tangerine Dream, Klaus Schulze and their followers): repeating step sequences that move with the chords and slowly mutate, bass lines, arpeggios, pads, drones and drum patterns. It is part of RetroSeq and writes its results straight to RetroSeq tracks, where you listen through your own VST3 instruments and MIDI synths, freeze what you like, edit it and export it.

This manual covers Kosmische in RetroSeq 0.21. RetroSeq itself is described in the *RetroSeq Reference Manual*.

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> Kosmische and RetroSeq are free. Donations are welcome via PayPal: **michael.ostman@gmail.com**

# 1. Introduction

## The Idea

A Berlin School piece is built from a few short patterns that repeat for a long time. The interest comes from small changes: a note that moves, an octave jump that comes and goes, the sequence following a slow chord change, an accent that shifts, a filter being turned by hand. Kosmische generates the patterns and the slow changes; you choose the sounds, listen, keep what works and play the rest.

Everything Kosmische generates comes from a **seed**, a number per track. The same settings and the same seed always give exactly the same notes, so a result you like can always be recreated. New seeds give new patterns in the same style.

## The Workflow

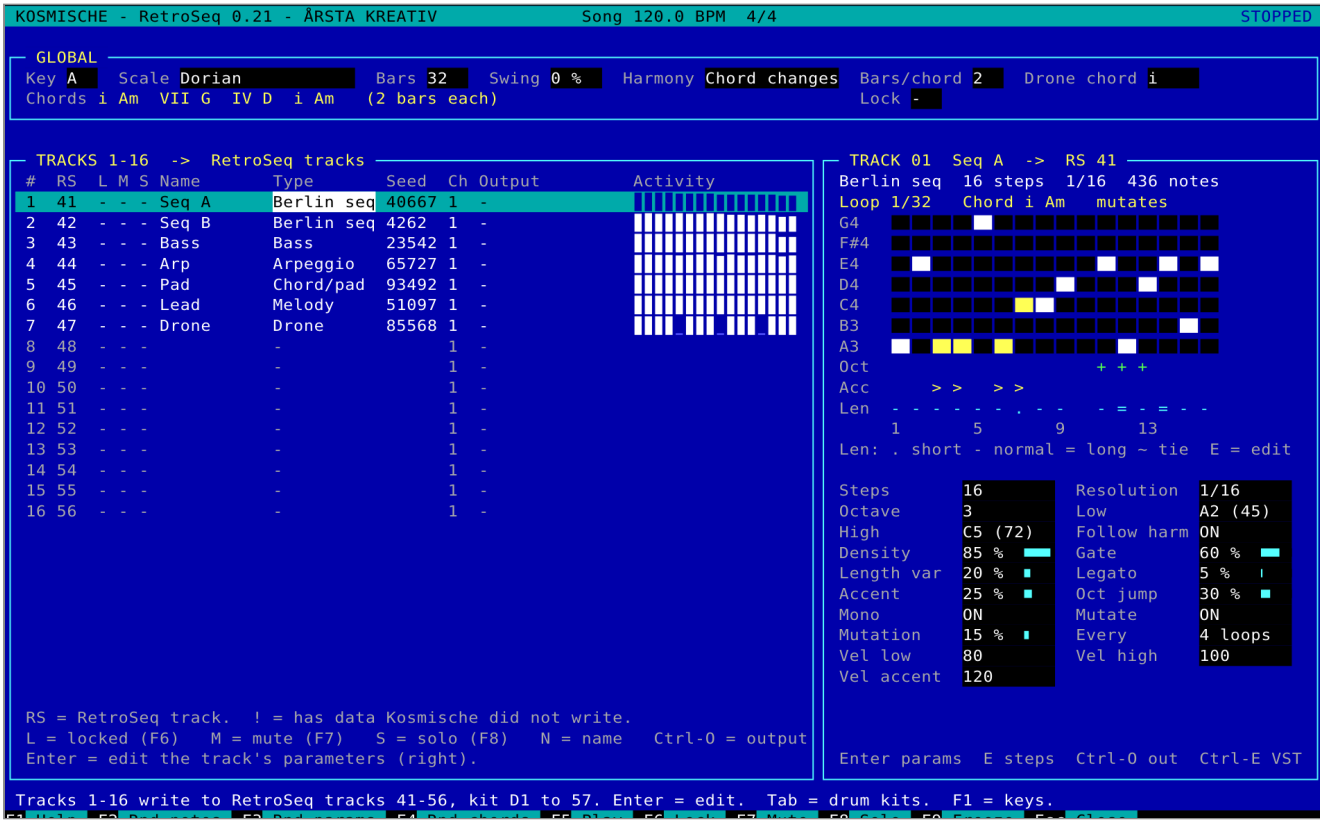
1. Open Kosmische with `K` from RetroSeq's Main or View screen.
2. Choose key, scale, length and chords (GLOBAL), the track types and their RetroSeq tracks.
3. Choose an output for each track: a VST3 instrument or a MIDI port and channel (`Ctrl-O`).
4. Press `Space` to listen to the loop. Press `F2` for new notes, `F3` for new parameters, `F4` for new chords. Lock (`F6`) what you want to keep.
5. Adjust parameters, or edit steps by hand (`E`).
6. Save the patch (`Ctrl-S`) so you can come back to it.
7. Freeze (`F9`) the tracks you are happy with. They become ordinary RetroSeq tracks.
8. Edit further in RetroSeq if you like, then **FILES → Export all** for one MIDI file per track.

## What Is Stored Where

| What | Where |
|---|---|
| Generated notes | In the linked RetroSeq tracks, and so in the RetroSeq song when you save it. |
| Kosmische settings: global, tracks, kits, lanes, seeds, locks, hand-edited steps, RS links | The Kosmische patch file (.kos), `Ctrl-S` / `Ctrl-L`. |
| Outputs, MIDI channels, VST3 instruments and their sounds, mute and solo | In the RetroSeq tracks and the RetroSeq song. Kosmische shows and sets them, but they belong to RetroSeq. |
| Tempo and time signature | The RetroSeq song. Kosmische follows them. |

Closing the Kosmische window only hides it; the settings stay until RetroSeq is closed. Save a patch to keep them.

# 2. The Kosmische Screen




The track view: GLOBAL at the top, the 16 tracks to the left and the editor for the selected track to the right.

## Opening and Closing

Press `K` in RetroSeq's Main or View screen. The first time, Kosmische generates all tracks and writes them to RetroSeq tracks 41–56 and the first drum kit to track 57. `Esc` closes the window (or goes back one level first). The window remembers its size and position and is always kept on the screen. Make it larger for larger text; the screen always has 128 × 40 character cells.

## Layout

| Area | Contents |
|---|---|
| Title row | Version, the RetroSeq song's tempo and time signature, and STOPPED or PLAYING with the current bar. |
| GLOBAL | Key, scale, length in bars, swing, harmony, chords and the chord lock (chapter 3). |
| Track list (left) | One row per track: **#**, **RS** (linked RetroSeq track), **L** lock, **M** mute, **S** solo, **Name**, **Type**, **Seed**, **Ch** MIDI channel, **Output** and **Activity** (notes per bar; the bar that is playing is lit). A * after the type means the pattern has been edited by hand. |
| Editor (right) | The selected track: a step grid (Berlin seq, Bass, Melody) or a piano roll (Arpeggio, Chord/pad, Drone), and all parameters for the track type. With GLOBAL selected, a short explanation. |
| Message row | Messages, questions (Y/N) and text entry. |
| Function key row | F1–F9 and Esc. The keys can also be clicked. |

`Tab` switches between the track view and the drum kit view (chapter 8). GLOBAL is the same in both.

## Moving Around and Changing Values

| Key | Action |
|---|---|
| `↑` `↓` | Select GLOBAL or a track. `PgUp` goes to GLOBAL, `PgDn` to the last row. |
| `←` `→` | Select a column. `Home` / `End` go to the first / last. |
| `+` `-` | Change the value by one step. `]` `[` change it by more. The mouse wheel works on the field under the pointer (with Ctrl: by more). |
| digits, `Enter` | Type a value in numeric fields (Bars, RS, Seed, Ch and parameters). |
| `Enter` | On a track: move into the editor's parameters (frame turns yellow); `Esc` goes back. On L, M, S, Name or Output: switch, rename or choose. |
| `N` | Rename the track (12 characters). The name is also given to the RetroSeq track. |
| Mouse | Click selects a field, double-click acts like Enter, the wheel changes values. |

# 3. Global Settings

| Field | Meaning |
|---|---|
| Key | Root note, C to B. `]` `[` move a fifth. |
| Scale | Major, Minor, Dorian, Phrygian, Lydian, Mixolydian, Harmonic minor, Pentatonic major, Pentatonic minor, Phrygian dominant, Hungarian minor. All tracks use the same key and scale. |
| Bars | The length that is generated and played in a loop. Mutation and chord changes happen within this length. |
| Swing | Moves every second step later, 0–50 % of a step. |
| Harmony | **Chord changes**: tracks follow a chord progression within the scale. **Drone chord**: one chord all the time; nothing is transposed. |
| Bars/chord | How many bars each chord lasts. |
| Drone chord | The scale degree of the chord in Drone chord mode. |
| Chords | The progression, shown as roman numerals and chord names (i Am, VII G …). `Enter` lets you type it as scale degrees, for example `1 7 4 1`. `+` / `-` or `F4` make a new one. |
| Lock | Locks the progression so that `F4` keeps it (also `F6` on GLOBAL). |

Tempo and time signature come from the RetroSeq song. If the time signature is changed in RetroSeq, Kosmische generates all tracks again.

# 4. Tracks

## Track Types

| Type | What it plays | Shown as |
|---|---|---|
| Berlin seq | The classic step sequence: a pattern of 1–64 steps (default 16 sixteenths) of chord tones and scale notes with octave jumps and accents. It loops, follows the chords and mutates slowly. | Step grid |
| Bass | A bass pattern built on root, fifth and third, by default 8 eighths, with more long and tied notes. | Step grid |
| Melody | A motif with notes of different lengths that mostly moves by step and lands on chord tones on strong beats. It is repeated and varied. | Step grid (long notes grey) |
| Arpeggio | The current chord broken up: up, down, up/down or random, over 1–4 octaves. | Piano roll |
| Chord/pad | The chord of each period, 2–5 notes. | Piano roll |
| Drone | Long notes: root, root + fifth, or root + fifth + octave. Repeated chords join into one long note. | Piano roll |
| - | Empty: the track is not used. | |

Change the type with `+` / `-` in the Type column. A new type gets its default parameters; the seed, the RetroSeq track and your own name are kept.

## RetroSeq Tracks (RS)

Every Kosmische track writes to one RetroSeq track, by default 41–56 for tracks 1–16 and 57–60 for drum kits D1–D4. Change it in the RS column (`+`/`-` or type a number). A RetroSeq track can only be used by one Kosmische track or kit. `.` or `Del` in the RS column freezes the track (chapter 9).

## Output, Channel, Mute and Solo

- **Output** (`Ctrl-O` or `Enter` in the Output column): a list of *No output*, the MIDI output ports and the VST3 instruments that RetroSeq knows. If the list has no instruments, scan for plug-ins in RetroSeq (`F4` there). This is the same setting as RetroSeq's Output column.
- `Ctrl-E` opens the VST3 instrument's own editor.
- **Ch**: the MIDI channel of the RetroSeq track (1–16). Drum kits use channel 10 when their RetroSeq track is empty.
- **M** and **S** (`F7`, `F8`) are RetroSeq's mute and solo for the track.

# 5. Track Parameters

Press `Enter` on a track to move into its parameters. Arrow keys select, `+` `-` `[` `]` or digits change. The track is generated again and written to RetroSeq at once, also while it plays.

| Parameter | Types | Meaning |
|---|---|---|
| Steps | Berlin, Bass, Melody | Length of the pattern in steps (1–64; Melody 2–128). Odd lengths (12, 7, 5) against the bar give the shifting Berlin School feel. |
| Resolution | all but Chord/pad, Drone | Step length: 1/4, 1/8, 1/8T, 1/16, 1/16T, 1/32. |
| Octave | all | The octave of the root (C4 = middle C = 60). |
| Low / High | all | Note range. Notes outside are moved in by whole octaves. |
| Follow harm | all | ON: transposed with the chord progression. OFF: always on the key's root chord. |
| Density | all | The chance that a step has a note (Arpeggio, Chord/pad: that a note or chord is played). |
| Gate | all | Note length as a part of the step (Chord/pad, Drone: of the chord period). |
| Length var | Berlin, Bass, Melody, Arpeggio | Share of steps that become Short or Long instead of Normal. |
| Legato | Berlin, Bass, Melody, Arpeggio | Share of steps that are tied to the next note. |
| Accent | Berlin, Bass, Melody, Arpeggio | Chance of an accent (velocity Vel accent). |
| Oct jump | Berlin, Bass | Chance that a step jumps an octave up. |
| Stepwise | Melody | Chance that the melody moves to a neighbouring note rather than leaps. |
| Arp mode | Arpeggio | Up, Down, Up/down, Random. |
| Arp octaves | Arpeggio | 1–4 octaves. |
| Chord notes | Arpeggio, Chord/pad | Notes in the chord: 3 = triad, 4 = seventh chord, 5 = ninth. |
| Drone notes | Drone | 1 = root, 2 = + fifth, 3 = + octave. |
| Mono | all | Two notes never sound at the same time (except the short overlap of a tie). For monophonic synths. |
| Mutate | Berlin, Bass, Melody | ON: the pattern changes slowly over the bars. |
| Mutation | Berlin, Bass, Melody | Share of the steps that change at each mutation. |
| Every | Berlin, Bass, Melody | Mutate every N loops of the pattern. |
| Vel low / Vel high | all | Velocity range for normal notes. |
| Vel accent | all with accents | Velocity of accented notes. |
| Seed | all (track list) | The number all randomness comes from. Type a seed to get a result back. |

## Note Lengths, Legato and Mono

Each step has a length class, shown in the Len row of the step grid:

| Symbol | Class | Length |
|---|---|---|
| . | Short | About half the gate: staccato. |
| - | Normal | The track's gate. |
| = | Long | The whole step; the next note gets a new attack. |
| ~ | Tie | Lasts until the next note plus 10 ticks of overlap, so a monophonic synth (or a VST in legato mode) glides into it without a new attack. A tie to the same pitch becomes one long note. A tie before a rest becomes Long. |

Defaults: Berlin seq Mono on, Length var 20 %, Legato 5 %. Bass Mono on, 30 %, 30 %. Melody Mono on, 0 %, 30 %. Arpeggio Mono on, no variation. Chord/pad and Drone are polyphonic.

# 6. Randomizing, Locks and Mutation

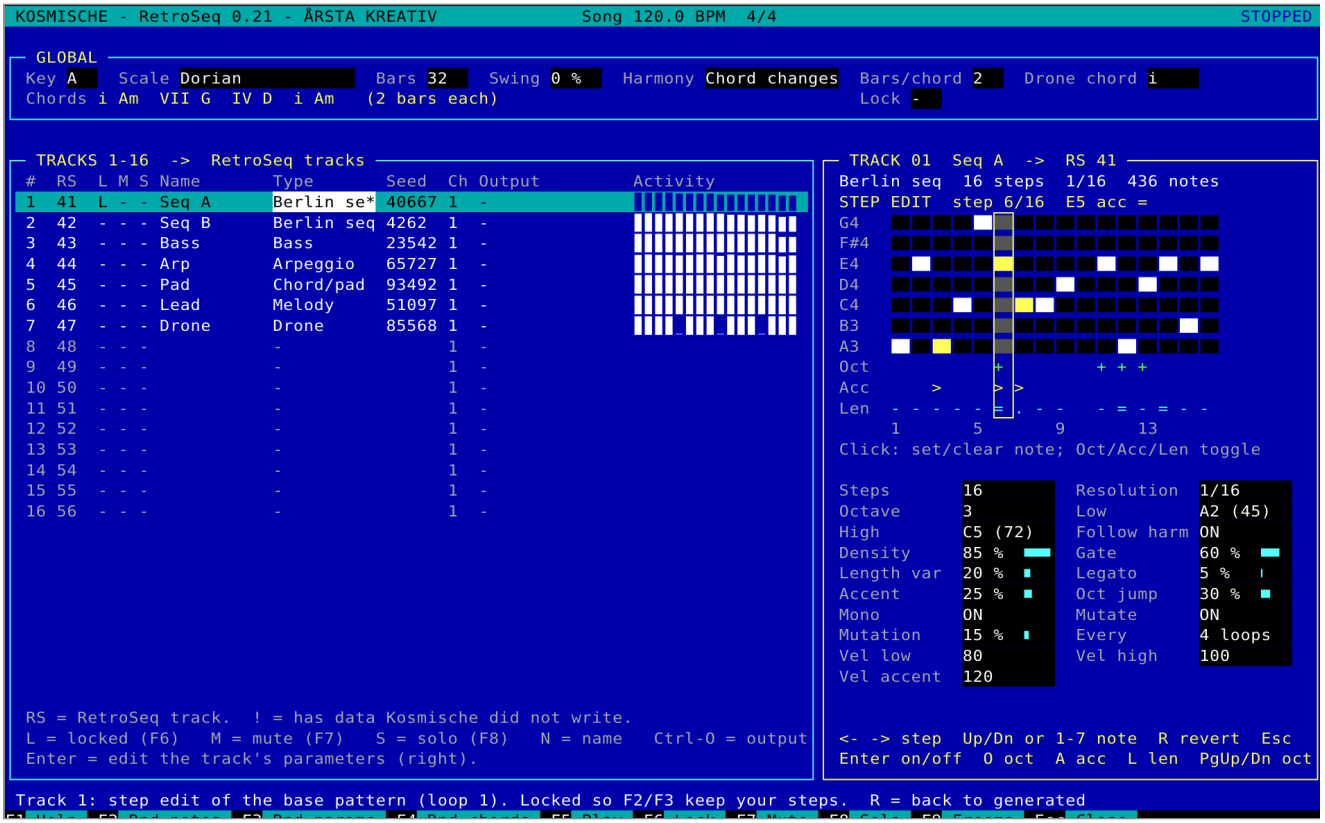
| Key | Action |
|---|---|
| F2 | **New notes**: new seeds for all unlocked tracks, kits and lanes. The parameters stay. |
| F3 | **New parameters** within sensible ranges for each type, for unlocked tracks, kits and lanes. Type, octave, range and seed stay. |
| F4 | **New chord progression**, unless it is locked. |
| F6 | Lock or unlock the selected track (on GLOBAL: the chords; in the kit view: the kit or lane). Locked tracks are marked L and are left alone by F2 and F3. |

A lock on one track does not change what the other tracks get: the same F2 gives the others the same new seeds whether a track is locked or not.

## Mutation

With Mutate on, the pattern is changed a little every N loops (Every): a step gets a new note, an octave jump is added or removed, a step is switched on or off, or a note length changes. Mutation is part of the generated notes, so the whole length (Bars) is fixed and plays the same every time. The step grid shows the pattern of the loop that is playing, with "Loop 3/32" above it, so you can watch the sequence develop.

# 7. Step Editing




Step edit of a Berlin seq track. The yellow frame is the cursor; the info row shows the step's note and length.

The pattern of a Berlin seq, Bass or Melody track can be edited by hand. Select the track and press `E`, or click a cell in the step grid. The grid then shows the **base pattern** (loop 1) and the frame turns yellow.

The first edit copies the generated pattern, so you start from what you hear. The track is locked at the same time, so `F2` and `F3` leave your steps alone. Harmony, range and velocities still apply. If Mutate is on, the mutation starts from your pattern; turn Mutate off to hear exactly what you edited in every loop.

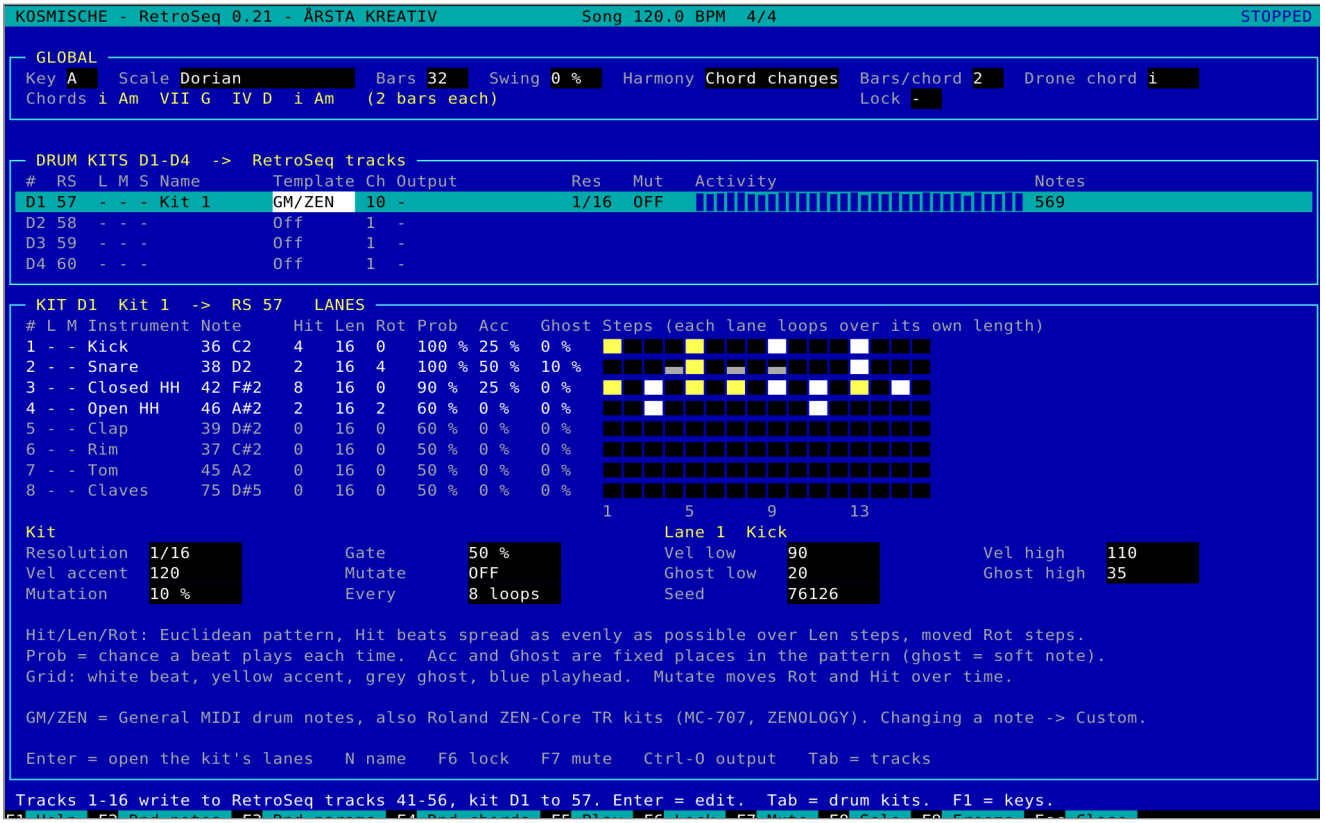
| Key | Action |
|---|---|
| `←` `→` | Move between steps (Ctrl: four steps). `Home` / `End`: first / last step. |
| `↑` `↓`, `+` `-` | Note one scale step up / down. On a rest the first press switches the note on. |
| `1` – `7` | Set the note to that degree of the current chord (1 = root, 3 = third, 5 = fifth), in the same octave. |
| `PgUp` `PgDn` | Note an octave up / down. |
| `Enter` / `Ins` | Switch the step on / off (the note is remembered). |
| `Del` `0` `.` | Make the step a rest. |
| `O` | Octave jump on / off (the + in the Oct row). |
| `A` | Accent on / off. |
| `L` | Next length: . short → - normal → = long → ~ tie. |
| `]` `[` | Melody only: the note lasts one step more / less (it always ends at the next note). |
| `R` | Revert: drop the hand-edited steps and go back to the generated pattern (asks Y/N). |

| | |
|---|---|
| `Esc` | Leave step edit. The edits stay. |
| Mouse | Click a cell to put the note there; click the note again to remove it. Click in the Oct, Acc or Len row to toggle or step it. |

The notes are degrees relative to the chord, so an edited pattern still follows the chord progression. Edited tracks are marked with ***** after the type and "edited" above the grid. Changing Steps keeps the edited steps (new steps are rests). Changing the type of a track drops the edits. The edits are saved in the .kos file.

Arpeggio, Chord/pad and Drone are not step-based and cannot be edited this way. Edit their notes in RetroSeq after Freeze.

# 8. Drum Kits




The kit view (Tab): the four kits at the top, the eight lanes of the selected kit below.

Press `Tab` for the drum kits. There are four kits, D1–D4, each with eight **lanes**; a lane is one instrument (one MIDI note) with its own pattern. A kit writes to one RetroSeq track, normally on channel 10, and is exported as one MIDI file.

## The Kit List

Columns: **#**, **RS**, **L**, **M**, **S**, **Name**, **Template**, **Ch**, **Output**, **Res** (resolution), **Mut** (mutation on/off), Activity and the number of notes. They work as in the track list. `Enter` on a kit opens its lanes; `↑` from lane 1 goes back to the kit.

## Templates and Drum Notes

| Template | Meaning |
|---|---|
| Off | The kit is not used. D2–D4 start as Off. |
| GM/ZEN | General MIDI drum notes, which Roland's ZEN-Core TR kits also use (MC-707, Fantom, ZENOLOGY; checked against the FANTOM-06/07/08 Sound List, Drum Kit Key Assign List). Switching from Off gives a ready kit with kick, snare and hi-hats. |
| Custom | Your own notes. Changing a lane's note in a GM/ZEN kit makes it Custom. |

| Lane | Instrument | Note | Default pattern |
|---|---|---|---|
| 1 | Kick | 36 (C2) | 4 hits / 16, on the beats |
| 2 | Snare | 38 (D2) | 2 hits, rotation 4 (beats 2 and 4), accent 50 %, ghost 10 % |
| 3 | Closed HH | 42 (F#2) | 8 hits, probability 90 % |
| 4 | Open HH | 46 (A#2) | 2 hits, rotation 2, probability 60 % |
| 5 | Clap | 39 (D#2) | off (0 hits) |
| 6 | Rim | 37 (C#2) | off |
| 7 | Tom | 45 (A2) | off |
| 8 | Claves | 75 (D#5) | off |

Other GM/ZEN notes: 49 crash cymbal, 56 cowbell. In the acoustic ZEN-Core kits note 39 is a ghost snare instead of a clap.

## Lanes and Euclidean Patterns

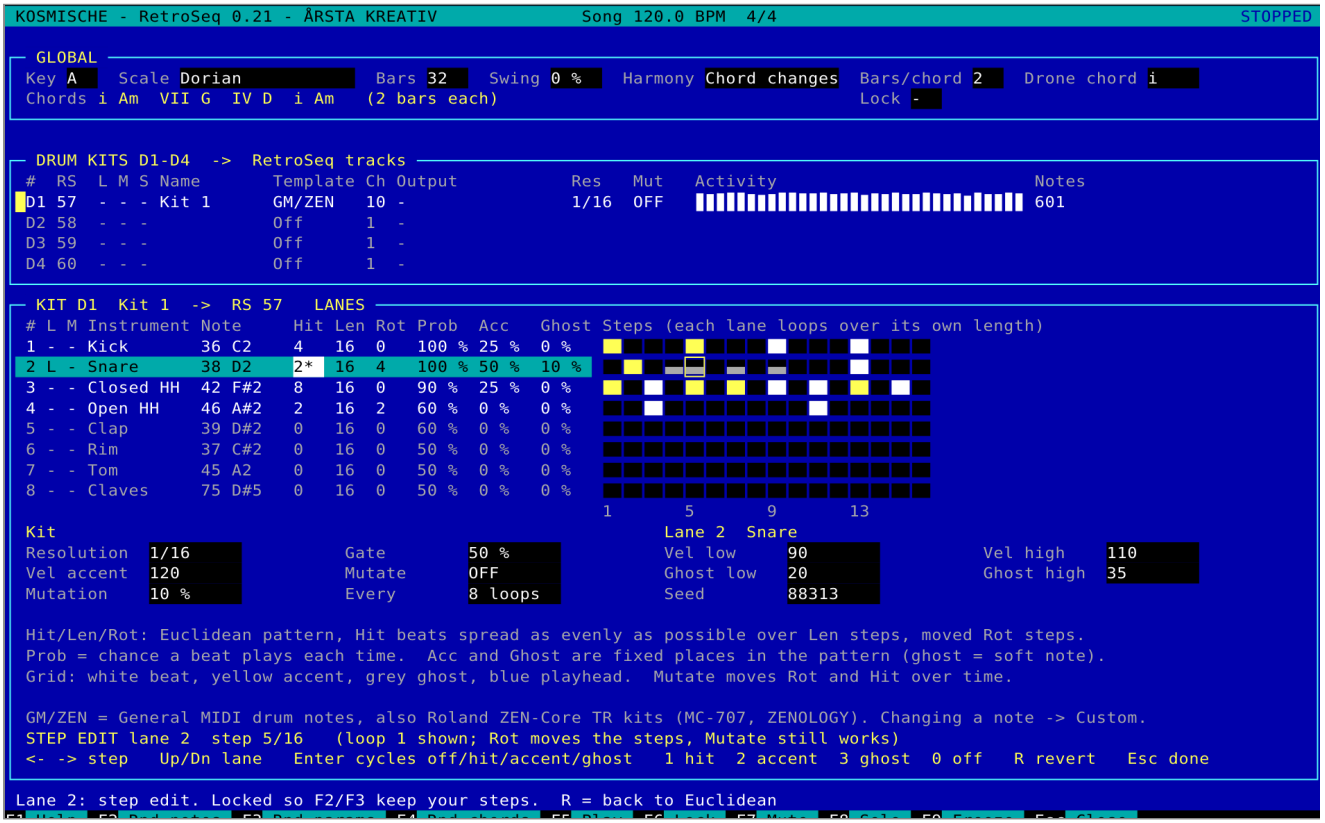
Each lane spreads **Hit** beats as evenly as possible over **Len** steps and moves the pattern **Rot** steps. 4 over 16 gives four on the floor; 3 over 8 gives the familiar x..x..x. figure; 5 over 16 and 7 over 16 give the rolling patterns of electronic music. Lanes can have different lengths: a 12-step lane against 16-step lanes gives a polyrhythm that comes round again every 48 steps.

| Column | Meaning |
|---|---|
| L, M | Lock (F2/F3 leave the lane alone) and mute of the lane (`F7` on a lane). |
| Instrument | The lane's name (`N`). |
| Note | The MIDI note. `]` `[` change an octave. |
| Hit, Len, Rot | The Euclidean pattern: beats, length in steps (1–64), rotation. |
| Prob | The chance that a beat is played, decided for every beat, so the pattern breathes. |
| Acc | Share of the beats that are accents (fixed places in the pattern; the first beat is always an accent when Acc > 0). |
| Ghost | Share of the empty steps that get a soft ghost note. |

Kit parameters (below the lanes, left): **Resolution** (1/8, 1/8T, 1/16, 1/16T, 1/32), **Gate**, **Vel accent**, **Mutate**, **Mutation**, **Every**. Mutation moves the rotation or changes the number of beats, on each lane's own cycle. Lane parameters (right): **Vel low / high**, **Ghost low / high** and the lane's **Seed**.

The grid shows each lane for the loop that is playing: white beat, yellow accent, grey (short) ghost, blue playhead.

# Editing a Lane by Hand




Step edit of lane 2. The snare got an accent on step 2 and a ghost on step 5; Hit shows 2* (edited).

Select a lane and press `E`, or click a cell in its grid. As with tracks, the first edit copies the current pattern and locks the lane.

| Key | Action |
|---|---|
| `←` `→` | Move between steps. |
| `↑` `↓` | Move to the lane above / below and keep editing. |
| `Enter`, `+`, click | Cycle the step: off → hit → accent → ghost → off. |
| `1` / `H` | Hit. |
| `2` / `A` | Accent. |
| `3` / `G` | Ghost. |
| `0` `Del` `.` | Off. |
| `R` | Revert to the Euclidean pattern (asks Y/N). |
| `Esc` | Leave step edit. |

An edited lane shows ***** after the number of hits. Rot still moves the edited pattern, Len makes it longer (new steps are off) or shorter, and Prob and Mutate still work (mutation only rotates an edited lane). Changing **Hit** replaces the edited steps with a Euclidean pattern again.

# 9. Listening, Freeze and Protection

## Listening

`Space` or `F5` plays bars 1 to Bars in a loop through RetroSeq, with all RetroSeq tracks that have notes; press again to stop. Changes are written to RetroSeq at once and heard in the next notes, also while playing. Mute and solo (`F7`, `F8`) work as in RetroSeq.

## Freeze

`F9` freezes the selected track or kit: the link is released and the notes stay on the RetroSeq track as ordinary notes. Kosmische never touches that RetroSeq track again. The RS field shows `--` until you choose a new RetroSeq track, and then the Kosmische track goes on generating there. Use Freeze for everything you want to keep and edit in RetroSeq.

## Protection

Kosmische only writes to RetroSeq tracks it has written itself. If a linked RetroSeq track has other notes (you edited it in RetroSeq, or it had notes from the start), Kosmische asks before overwriting: *Overwrite? Y = yes, N = skip*. Skipped tracks are marked with **!** in the RS column and are not written until you confirm or choose another RetroSeq track. When a track is taken over, RetroSeq's Trans and Qnt for it are reset.

When a song is loaded or a new song is started in RetroSeq, Kosmische forgets which tracks it has written; the links stay, and Kosmische asks before writing to a linked track that has notes.

# 10. Files

## The Kosmische Patch (.kos)

| Key | Action |
|---|---|
| `Ctrl-S` | Save the patch. The first time you choose a file name; after that it saves to the same file. |
| `Ctrl-Shift-S` | Save as a new file. |
| `Ctrl-L` | Load a patch. All tracks and kits are generated again and written to their RetroSeq tracks. |

The .kos file contains the global settings and chord progression, all 16 tracks and 4 kits with their parameters, seeds, locks, hand-edited steps and lanes, and the links to RetroSeq tracks. It is a readable XML file. It does not contain outputs, VST3 sounds, mute or solo; they are saved in the RetroSeq song. Save both the song (RetroSeq FILES → Save) and the patch (`Ctrl-S`) to come back to exactly the same session.

When a patch is loaded, tracks that keep their RetroSeq track are written without questions; if a track moves to another RetroSeq track, the old one keeps its notes (as after Freeze), and Kosmische asks before writing over notes it did not write.

## Export All

In RetroSeq: **FILES → Export all**. Every RetroSeq track with notes is saved as its own MIDI file (type 0) in a folder you choose, named after the track number and name, for example `41 Seq A.mid` or `57 Kit 1.mid`. This is how Kosmische tracks reach your DAW, one file per instrument.

- If any track has Trans, Qnt, Vol, Prg, Pan or velocity scaling set, RetroSeq asks once: **Y** exports *as played* (with those settings applied), **N** exports the raw notes.
- If some of the files already exist in the folder, RetroSeq asks once before overwriting.
- The folder is remembered for the next export.

Export all exports what is in the RetroSeq tracks, frozen or still linked, together with the rest of the song.

# 11. Quick Reference Card

| Key | Action |
|---|---|
| `K` (in RetroSeq) | Open Kosmische |
| `Esc` | Back one level / close Kosmische |
| `Tab` | Tracks / drum kits |
| `↑` `↓` `←` `→` `Home` `End` `PgUp` `PgDn` | Select row and column |
| `+` `-` `]` `[`, wheel | Change value (by one / by more) |
| digits `Enter` | Type a value |
| `Enter` | Parameters / switch / rename / choose; on a kit: its lanes |
| `N` | Rename track, kit or lane |
| `E` | Step edit (track pattern or drum lane) |
| `Ctrl-O` | Choose output (MIDI port or VST3) |
| `Ctrl-E` | Open the VST3 editor |
| `Ctrl-S` / `Ctrl-Shift-S` / `Ctrl-L` | Save / save as / load patch (.kos) |
| `F1` | Keys |
| `F2` | New notes (seeds) for unlocked tracks, kits and lanes |
| `F3` | New parameters for unlocked tracks, kits and lanes |
| `F4` | New chord progression |
| `F5` / `Space` | Play loop / stop |
| `F6` `F7` `F8` | Lock / mute / solo |
| `F9` | Freeze |
| `Y` / `N` | Answer a question on the message row |

**Step edit:** `←` `→` step · `↑` `↓` or `1`–`7` note · `PgUp` `PgDn` octave · `Enter` on/off · `0` octave jump · `A` accent · `L` length · `]` `[` hold (Melody) · `R` revert · `Esc` done.
**Lane edit:** `Enter` cycles off/hit/accent/ghost · `1` hit · `2` accent · `3` ghost · `0` off · `↑` `↓` lane.

# 12. Version History

| Version | Kosmische |
|---|---|
| 0.17 | First version in RetroSeq: generator for Berlin seq, Bass, Melody, Arpeggio, Chord/pad and Drone; tracks written to RetroSeq tracks; listening in a loop; Freeze. |
| 0.18 | Version and ÅRSTA KREATIV in all titles; keyboard focus between the windows. |
| 0.19 | The main screen: track list and editor with step grid and piano roll following the playhead, all parameters, names, channels, outputs, mute and solo, typed chord progressions, swing. |
| 0.20 | Drum kits D1–D4 with eight Euclidean lanes each, GM/ZEN template, channel 10, kit view (Tab). |
| 0.21 | Step editing of tracks and lanes; the .kos patch file; Export all in RetroSeq; this manual. |

> Kosmische and RetroSeq are free. Donations are welcome via PayPal: **michael.ostman@gmail.com**
>
> RetroSeq 0.21 - ÅRSTA KREATIV